

Effective-temperature multipoles for relativistic radiation transport.

IV. Observable semantics and reconstructive adequacy

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The effective-temperature (Teff) multipole representation parameterises angular nonequilibrium in relativistic kinetic theory by a direction-dependent temperature field $\Theta(\hat{e})$ on the observer’s local momentum sphere. Papers I–III of this series developed the tangency diagnostic, chemical-potential extension, and entropy geometry of the Teff manifold; in all three, the reference temperature T_0 served as an internal gauge-fixing device whose relationship to observer-measurable quantities was not made explicit. We fill this gap. For blackbody photons (Planck distribution with $\mu = 0$), $T_{\text{eff}}(\hat{e}) = T_0\Theta(\hat{e})$ is the exact brightness temperature at all frequencies; for Fermi–Dirac and Bose–Einstein statistics it is the occupation temperature coinciding with the brightness temperature only in the Wien tail. The bolometric temperature T_{bol} exceeds T_0 by a statistics-independent $O(A^2)$ correction governed by the angular variance of Θ (Theorem 1). The nonlinear moment map connecting Teff multipoles to energy-density moments is locally invertible at generic configurations (Theorem 2). We derive a conditional local bound connecting the off-manifold state residual to the observable reconstruction error (Theorem 3): $\Delta_{\text{obs}} \leq C_{\text{loc}} \mathcal{D}_{\geq 2}^{\text{state}} / \sigma_{\min}(J)$ under explicit assumptions (H1–H3), where C_{loc} is a sample-dependent local calibration constant. A matched-truncation-order comparison with five production closures—Levermore M_1 , Minerbo, MEFD, PSTF, and Teff—shows comparable observable fidelity overall, with Teff providing manifold-domain realizability ($f \geq 0$ wherever $\Theta > 0$) and a built-in validity diagnostic. Validation against BGK collision-generated spectra confirms the diagnostic bridge. We discuss implications for neutrino transport in binary neutron star mergers, CMB spectral distortions, and core-collapse supernovae.

I. INTRODUCTION

A. The closure problem and its observational consequences

The accuracy of radiation-transport simulations in relativistic astrophysics depends critically on the angular closure. In binary neutron star (BNS) merger simulations, the choice between M_1 closure and Monte Carlo transport produces 10–30% differences in global properties including neutrino luminosities, mean energies, and ejecta electron fraction Y_e [1, 2]. The M_1 closure forces the radiation field onto a single curve in the (f, χ) plane of flux factor and Eddington factor, incapable of representing beam-crossing configurations [3, 4]. In core-collapse supernovae (CCSN), the relationship between Eddington-factor errors and dynamical outcomes is non-monotonic [5], underscoring the need for closure diagnostics beyond pointwise accuracy metrics.

A distinct but related problem is spectral: energy-averaged thermalization of neutrinos does not guarantee accurate weak interaction rates. Alford et al. [6] demonstrate that bulk properties may agree with thermal assumptions while the distribution-function shape departs sufficiently to alter charged-current rates, with implications for Y_e and nucleosynthesis.

B. Existing closures and their limitations

The standard moment closures include the entropy-based M_1 of Levermore [7, 8] and Dubroca & Feugéas [9], the maximum-entropy Minerbo closure [10], and the PSTF truncation of Thorne [11]. A systematic comparison of seven M_1 -type closures by Murchikova et al. [12] quantified Eddington-factor differences but found no closure providing a diagnostic-guided error bound. Machine-learning surrogates—including ICNN-based closures [13] and gradient-boosting models [14]—improve accuracy but lack diagnostic-guided error certification. In the quantum-coherent sector, Froustey et al. [15] and Kneller et al. [16] derived the first analytic maximum-entropy closures for flavor-coherent neutrino transport; even there, no truncation diagnostic is available.

C. The Teff multipole representation

Papers I–III [17] of this series developed the effective-temperature multipole framework: a per-direction equilibrium ansatz $f_{\Theta} = \Phi_{\xi}(y/\Theta(\hat{e}))$ with gauge $\langle \Theta \rangle_{\Omega} = 1$, a tangency diagnostic $\mathcal{D}_{\geq 2}$ detecting collision-generated off-manifold content, an entropy-projection inequality, a direction-dependent chemical potential, and the entropy-geometric structure of the two-field manifold.

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D. Literature gaps and contributions

Three gaps define our contributions: (1) No diagnostic-guided observable error bound exists for truncated moment closures [18–20]. (2) The Gorban–Karlin invariant-manifold methodology has not been applied to radiation transport. (3) Covariant operational semantics for T_0 have not been developed.

We make eight contributions: **C1** (Covariant observable semantics, §II); **C2** (Bridge hierarchy, §II C); **C3** (Local identifiability, §III B); **C4** (Diagnostic-observable bridge, §IV); **C5** (Five-closure comparison, §V); **C6** (Neutrino application, §VII); **C7** (Self-consistent validation, §VI); **C8** (Explicit literature-gap claims).

The paper does not build a transport solver, address polarisation (Paper V), or prove global-in-time stability. Unless explicitly stated otherwise, all operational semantics and adequacy statements are formulated for the one-field scalar- η Teff family. The direction-dependent chemical-potential extension of Paper II changes the cross-sector observables already at first order.

II. COVARIANT FRAMEWORK AND OBSERVABLE SEMANTICS

A. Covariant setting

We work on $(\mathcal{M}, g_{ab})$ with signature $(-, +, +, +)$. A timelike observer u^a defines $h_{ab} := g_{ab} + u_a u_b$ [11, 21, 22]. A particle with p^a has $E := -u_a p^a > 0$ and spatial direction e^a with $p^a = E(u^a + v e^a)$ [23]. The Teff ansatz in this setting is

$$f_\Theta(x^a; E, e^a) = \Phi_\xi \left(\frac{E - \mu}{k_B T_0(x) \Theta(x, e^a)} \right). \quad (1)$$

Covariant entropy closures in curved spacetime were developed by Banach & Larecki [24]; the present work provides a per-direction equilibrium shape with the diagnostic $\mathcal{D}_{\geq 2}$ as validity diagnostic.

B. Occupation temperature versus brightness temperature

We call T_0 the *occupation temperature*. For blackbody photons (Planck distribution with $\mu = 0$), the specific intensity along direction $\hat{e}$ is $I_\nu(\hat{e}) = B_\nu(T_0 \Theta(\hat{e}))$, and the Planck-inversion brightness temperature is

$$T_b(\nu, \hat{e}) = T_0 \Theta(\hat{e}) \quad (\text{Planck, } \mu = 0, \text{ all } \nu). \quad (2)$$

This identification is exact and frequency-independent because the per-ray spectrum is a Planck function with a direction-dependent temperature. For general Maxwell–Boltzmann statistics with $\mu \neq 0$, the per-ray distribution $\Phi_\xi(y/\Theta)$ is not a Planck function, and the Planck-

inversion brightness temperature acquires a frequency-dependent offset from the chemical potential.

For FD the brightness temperature is frequency-dependent within the Teff manifold even at $\mu = 0$, because $\Phi_\xi(y/\Theta)|_{\xi=-1} = 1/(e^{y/\Theta} + 1)$ is not a Planck function. For BE at $\mu = 0$, the distribution $1/(e^{y/\Theta} - 1)$ is the Planck function, so the brightness temperature is exact—this is the blackbody photon case of Eq. (2). For BE at $\mu < 0$, the chemical potential breaks the Planck form and the brightness temperature becomes frequency-dependent. In the Wien tail, $T_b \rightarrow T_0 \Theta$ with exponential convergence; in the Rayleigh–Jeans limit, statistics-dependent corrections arise. For neutrinos ($E_\nu \sim 10$ MeV, $T \sim 5$ MeV), the dimensionless energy $y \sim 2$ places the distribution in neither the Wien nor Rayleigh–Jeans regime, making the occupation temperature the natural per-direction observable.

C. Temperature bridge hierarchy

Theorem 1 (Covariant temperature bridge hierarchy). *Let $\Theta(\hat{e}) > 0$ with $\langle \Theta \rangle_\Omega = 1$. Define $T_{\text{bol}}^4 := T_0^4 \langle \Theta^4 \rangle_\Omega$ and $T_{\text{num}}^3 := T_0^3 \langle \Theta^3 \rangle_\Omega$. Then:*

- (i) $T_{\text{bol}} > T_{\text{num}} > T_0$ for any nonisotropic Θ .
- (ii) The bolometric bridge satisfies

$$\frac{T_{\text{bol}}}{T_0} = \langle \Theta^4 \rangle_\Omega^{1/4} = 1 + \frac{3}{2} \langle \theta^2 \rangle_\Omega + \langle \theta^3 \rangle_\Omega + \mathcal{O}(\theta^4), \quad (3)$$

where $\theta := \Theta - 1$. For $L = 2$ axisymmetric, $\langle \theta^2 \rangle_\Omega = A^2/3 + 4Q^2/45$ and $\langle \theta^3 \rangle_\Omega = 4A^2Q/15 + 16Q^3/945$. The cubic term vanishes for pure dipole ($Q = 0$) by parity but is generically nonzero.

(iii) The bridge is statistics-independent: the Stefan–Boltzmann prefactor cancels in the ratio.

Proof. (i) By strict monotonicity of L^p means for non-constant positive Θ with $\langle \Theta \rangle_\Omega = 1$: $\langle \Theta^p \rangle_\Omega^{1/p}$ is strictly increasing in p , giving $T_{\text{bol}}^{1/4} > T_{\text{num}}^{1/3} > T_0$. (ii) Taylor expansion of $\langle (1 + \theta)^4 \rangle_\Omega$ to $\mathcal{O}(\theta^3)$, using $\langle \theta \rangle_\Omega = 0$. (iii) The spectral integral $c_\xi = \int_0^\infty z^3 \Phi_\xi(z) dz$ cancels in the ratio T_{bol}/T_0 . $\square$

D. Statistics independence of moment ratios

A structural consequence of the factored form (1) is that all energy-density moment ratios—flux factor $\mathfrak{f} = \langle \mu \Theta^4 \rangle / \langle \Theta^4 \rangle$, Eddington factor $\chi = \langle \mu^2 \Theta^4 \rangle / \langle \Theta^4 \rangle$, and higher—are independent of the statistics parameter ξ . The prefactor c_ξ cancels in every ratio. Cross-sector ratios such as $f_{\mathcal{N}}/f_{\mathcal{E}}$ are not statistics-independent, because $\langle \Theta^3 \rangle / \langle \Theta^4 \rangle$ depends on the anisotropy—the structural observation underlying the chemical-potential dipole extension of Paper II.

III. RECOVERY AND LOCAL IDENTIFIABILITY

A. Exact-family recovery

When the true distribution lies on the Teff manifold— $f_{\text{real}} = \Phi_{\xi}(y/\Theta_{\text{real}}(\hat{e}))$ with PSTF content $\ell \leq L_{\text{real}}$ —the truncated Teff recovery at order $L \geq L_{\text{real}}$ reproduces the multipoles and the distribution exactly, provided the nonlinear moment map is locally injective. This is a self-consistency check, not a deep result; the nontrivial content lies in the injectivity condition and in the characterisation of what happens when the true distribution departs from the manifold.

We have verified exact recovery numerically for pure-dipole ($L_{\text{real}} = 1$, A up to 0.80) and dipole-plus-quadrupole ($L_{\text{real}} = 2$, six (A, Q) pairs with $A \neq 0$) configurations across MB, FD, and BE statistics. In every case, the recovered multipoles agree with the true values to better than 10^{-16} , and all seven observables of §IV A are reproduced to machine precision (Table I).

B. Local identifiability

The moment-map Jacobian determines whether the Teff multipoles can be uniquely recovered from the matched moments.

Theorem 2 (Generic local identifiability). *The nonlinear moment map of the axisymmetric $L = 2$ Teff family is locally invertible on an open dense subset of the admissible parameter space. The parity symmetry $\Theta(\mu; A, Q) = \Theta(-\mu; -A, Q)$ organises the matched moments into even and odd sectors: the even moments ($\ell = 0, 2$) are invariant under $A \rightarrow -A$, while the odd moment ($\ell = 1$) changes sign. Hence the symmetry does not obstruct local invertibility when the odd matched moment is retained, although reduced even-only observable sets may suffer symmetry-induced degeneracies.*

Proof. The moment-map components $M_{\ell}(A, Q) = \langle \mu^{\ell} \Theta(A, Q)^4 \rangle$ ($\ell = 0, 1, 2$) are real-analytic functions of (A, Q) , because $\Theta(\mu; A, Q) = 1 + A\mu + Q(\mu^2 - 1/3)$ is polynomial in (A, Q) . The Gram determinant $\Delta(A, Q) := \det(J^T J)$ of the 3×2 Jacobian J is therefore real-analytic.

At the isotropic point $(A, Q) = (0, 0)$, where $\Theta \equiv 1$, the Jacobian evaluates to

$$J(0, 0) = \begin{pmatrix} 0 & 0 \\ 4/3 & 0 \\ 0 & 16/45 \end{pmatrix},$$

using $\partial_A M_{\ell}|_{(0,0)} = 4\langle \mu^{\ell+1} \rangle$ and $\partial_Q M_{\ell}|_{(0,0)} = 4\langle \mu^{\ell}(\mu^2 - 1/3) \rangle$. Hence $\Delta(0, 0) = \frac{16}{9} \cdot \frac{256}{2025} \approx 0.225 > 0$.

Since Δ is real-analytic and not identically zero, its zero set $\{\Delta = 0\}$ is a proper analytic subset of $\mathbb{R}^2$ with empty interior. The complement—on which $\Delta > 0$ and

the moment map is locally invertible—is open and dense. $\square$

Numerical conditioning. On the sampled axisymmetric $L = 2$ configurations used in this paper ($|A| \in [0.05, 0.80]$, $|Q| \in [0, 0.30]$, 200 points), $\sigma_{\min}(J) > 10^{-2}$ and $\sigma_{\max}(J) < 10$, giving $\kappa(J) < 10^3$. The condition number is moderate at all sampled configurations.

C. Off-family behaviour: two regimes

Seven distortion types are tested: Compton y , chemical-potential shift, quadratic energy distortion, Pauli depletion, multi-beam, inelastic, and BGK-generated. The test suite spans 1008 configurations (two statistics $\times$ eight dipole amplitudes $\times$ four quadrupole amplitudes $\times$ three distortion amplitudes per type).

Two empirical regimes emerge (Table II).

Regime I (742 points, $\langle \Delta_{\text{obs}}/\Delta_{\text{ptw}} \rangle = 0.72$): spectral distortions exciting predominantly the $n \leq 1$ Laguerre modes. The per-ray temperature and normalisation updates absorb the smooth spectral shift; the residual is invisible to energy-integrated observables. The chemical-potential shift is the most dramatic example: $\Delta_{\text{obs}}/\Delta_{\text{ptw}} = 0.018$ ($\tilde{\mu} = 0.01$)—the observable error is 55 times smaller than the pointwise distribution error.

Regime II (266 points, $\langle \Delta_{\text{obs}}/\Delta_{\text{ptw}} \rangle = 1.94$): energy-shape distortions exciting $n \geq 2$ modes. The moment-matching optimiser absorbs the spectral content into shifted angular parameters, creating cross-contamination: at $\varepsilon = 0.02$ (quadratic distortion), the recovered quadrupole $Q_{\text{rec}} = 0.44$ versus the true $Q_{\text{true}} = 0.12$. Observable errors then exceed pointwise errors by a factor of 2.7–3.9.

The tangency diagnostic $\mathcal{D}_{\geq 2}$ discriminates between these regimes and provides the quantitative bound of Theorem 3.

IV. ADEQUACY CRITERION WITH DIAGNOSTIC BRIDGE

A. Observable set

The adequacy question requires specifying *against which observables* adequacy is judged. We assess recovery against seven observables (O_1 – O_7): three energy-density moments $\mathcal{E}_0$ (monopole), $\mathcal{F}_1$ (dipole flux), $\mathcal{P}_2$ (Eddington pressure), matched by the Teff recovery; two number-density moments $\mathcal{N}_0$ (monopole) and $\mathcal{N}_1$ (dipole flux), predicted but not matched; and two spectral diagnostics: the mean energy $\langle E \rangle = \mathcal{E}_0/\mathcal{N}_0$ and the RMS energy $\langle E^2 \rangle^{1/2}$.

The mean energy and its square are critical for neutrino physics: the charged-current cross-section scales as $\sigma \propto E_{\nu}^2$ [25], so a systematic bias in the recovered

energy distribution propagates quadratically into weak-interaction rates. The observable error is defined as

$$\Delta_{\text{obs}} := \max_{k=1,\dots,7} \frac{|O_k^{\text{rec}} - O_k^{\text{true}}|}{|O_k^{\text{true}}|}, \quad (4)$$

the maximum relative error over the seven-element observable set. The dominant contribution typically comes from the number-sector predictions (O_4, O_5), because the Θ^3 weighting probes a different angular average than the Θ^4 weighting used for moment-matching.

B. Laguerre decomposition of off-manifold content

The key to connecting the tangency diagnostic to the observable error is the spectral decomposition of the off-manifold perturbation. Let the true distribution be $f = f_{\Theta} + \delta f$, where f_{Θ} is the on-manifold component and δf is the off-manifold perturbation. Expanding δf in the generalised Laguerre basis adapted to the weight w_* :

$$\delta f(x, \hat{e}) = f(1+\xi f) \sum_{n=0}^{\infty} c_n(\hat{e}) L_n^{(\alpha_{\xi})} \left(\frac{x}{\Theta(\hat{e})} \right), \quad (5)$$

where α_{ξ} is the Laguerre index matching the spectral weight (Paper I, Sec. V A). The $n = 0$ mode shifts the amplitude (absorbable by η -update); the $n = 1$ mode shifts the spectral slope (absorbable by Θ -update); the $n \geq 2$ modes are the *genuine* spectral distortions that no per-ray temperature or chemical potential adjustment can absorb.

The tangency diagnostic of Paper I is

$$\mathcal{D}_{\geq 2} = \|c_{n \geq 2}\|_* = \sqrt{\sum_{n \geq 2} \langle c_n, c_n \rangle_*}, \quad (6)$$

the norm of the $n \geq 2$ Laguerre content of the collision source G_{ξ} .

C. The error bound

Theorem 3 (Conditional local diagnostic–observable bridge). *Assume: (H1) the state-side off-manifold residual (the $n \geq 2$ Laguerre content of the current distribution) is controlled by the Paper I collision-side tangency diagnostic—concretely, $\|c_{n \geq 2}\|_w \leq C_{\text{dyn}} \mathcal{D}_{\geq 2}^{\text{coll}}$ for a local constant C_{dyn} (this is the bridge between Paper I’s gap-opening-rate diagnostic and the current state; it is an assumption, not a consequence of Paper I); (H2) the recovery Jacobian J of Theorem 2 has full column rank ($\sigma_{\min}(J) > 0$); (H3) the observable map is C^1 in the recovery coordinates. Let the off-manifold perturbation have Laguerre decomposition (5) with $n \geq 2$ content of norm $\mathcal{D}_{\geq 2}^{\text{state}} := \|c_{n \geq 2}\|_w$. Then:*

(i) *The $n \leq 1$ components are absorbed by per-ray parameter updates: $(\Theta, \eta) \rightarrow (\Theta + \delta\Theta, \eta + \delta\eta)$ with*

$\delta\Theta = -c_1(\hat{e}) \Theta^2$ and $\delta\eta = -c_0(\hat{e})$. The energy-integrated residual depends only on $n \geq 2$ content.

(ii) *The angular parameter shift satisfies*

$$\|\delta T\|_2 \leq \frac{K}{\sigma_{\min}(J)} \mathcal{D}_{\geq 2}^{\text{state}}, \quad (7)$$

where δT is the shift in the Teff multipoles from the moment-matching inversion and K depends on the spectral weight and angular quadrature.

(iii) *The observable reconstruction error satisfies*

$$\Delta_{\text{obs}} \leq \frac{C_{\text{loc}}}{\sigma_{\min}(J)} \mathcal{D}_{\geq 2}^{\text{state}} + \mathcal{O}((\mathcal{D}_{\geq 2}^{\text{state}})^2), \quad (8)$$

where C_{loc} absorbs K and the observable Lipschitz constant.

(iv) *The local constant C_{loc} depends on (ξ, L, Θ, η) and is not a universal analytic constant.*

Remark 4 (Numerical calibration). At $L = 2$, scanning over 1008 configurations, the supremum of $\Delta_{\text{obs}} \cdot \sigma_{\min}(J) / \mathcal{D}_{\geq 2}$ yields $C_{\text{loc}}(\text{MB}, 2) \approx 3.2$ and $C_{\text{loc}}(\text{FD}, 2) \approx 4.1$. These are calibration values determined numerically over the sampled set; they are not universal analytic constants.

Proof. Step 1: per-ray affine absorption. The $n = 0$ and $n = 1$ Laguerre modes span the affine per-ray tangent space $V_{\text{aff}} = \text{span}\{v_1, v_2\}$ generated by the spectral-slope direction $v_1 = x/\Theta^2$ ($n = 1$) and the amplitude-shift direction $v_2 = 1$ ($n = 0$). A per-ray affine update $\Theta \rightarrow \Theta - c_1 \Theta^2$, supplemented by an amplitude rescaling absorbing c_0 , shifts the on-manifold distribution by exactly $f(1+\xi f)(c_0 + c_1 x/\Theta^2)$, cancelling the $n \leq 1$ components of δf . The residual after this update has only $n \geq 2$ content. (This affine tangent space coincides with the two-field tangent space of Paper III; the present argument uses only the per-ray spectral structure, not the angular $\eta(\hat{e})$ extension.)

Step 2: moment-matching perturbation. The energy-integrated moments respond to the residual through the moment-map Jacobian J . At linear order, the perturbation in the matched moments $(\mathcal{E}_0, \mathcal{F}_1, \mathcal{P}_2)$ from the $n \geq 2$ content is a vector $\delta \mathbf{m}$ satisfying $\|\delta \mathbf{m}\| \leq K \|c_{n \geq 2}\|_w$ for a constant K depending on the spectral weight and angular quadrature. The moment-matching inversion recovers multipoles $T + \delta T$ where $J \delta T = \delta \mathbf{m}$, so $\|\delta T\| \leq \|\delta \mathbf{m}\| / \sigma_{\min}(J) \leq K \mathcal{D}_{\geq 2} / \sigma_{\min}(J)$.

Step 3: observable propagation. The shifted multipoles $T + \delta T$ produce a reconstructed distribution whose observables differ from the true values by $|\delta O_k| \leq C_k \|\delta T\|$ for each observable, where C_k incorporates the sensitivity of O_k to the angular parameters. Taking the maximum over $k = 1, \dots, 7$ and setting $C = \max_k C_k \cdot K$ yields (8).

Step 4: constant determination. The constants C_{loc} are determined numerically by scanning over 1008 configurations (8 dipole amplitudes $\times$ 4 quadrupole amplitudes $\times$ 7 distortion types $\times$ 3 amplitudes per type $\times$ 2 statistics) and computing the ratio $\Delta_{\text{obs}} \cdot \sigma_{\min}(J) / \mathcal{D}_{\geq 2}$. The

reported values $C_{\text{loc}}(\text{MB}, 2) \approx 3.2$ and $C_{\text{loc}}(\text{FD}, 2) \approx 4.1$ are the suprema over the scanned configurations. $\square$

Remark 5 (Local versus dynamical adequacy). Theorem 3 bounds the *local* reconstruction error at a single spacetime point. Wang & Burrows [5] showed that Eddington-factor errors and dynamical outcomes are nonmonotonically related in CCSN: a locally adequate closure may produce globally inadequate dynamics (and vice versa). The bound (8) is a necessary condition for dynamical adequacy—if local reconstruction is inadequate, the time-integrated outcome is certainly corrupted—but not sufficient.

Practical consequence. The Teff recovery of mean energy $O_6 = \mathcal{E}_0/N_0$ provides a direct proxy for neutrino reaction rate accuracy. At $\mathcal{D}_{\geq 2} < 0.01$, the bound (8) gives $\Delta_{\text{obs}} \lesssim 0.04$ (taking $\sigma_{\text{min}} \approx 1$ at generic configurations)—mean energy error at the percent level. At $\mathcal{D}_{\geq 2} > 0.05$, Δ_{obs} can reach several percent, entering the regime where weak-rate accuracy degrades. The diagnostic thus provides a computable threshold for switching from the Teff closure to a higher-fidelity method.

D. Distortion-conditional ordering

The three fidelity levels—observable (Δ_{obs}), functional (Δ_{func} , weighted L^2 norm), and pointwise (Δ_{ptw} , maximum relative error over the (y, μ) grid)—do not admit a universal ordering.

Regime I (spectral distortions: $n \leq 1$ dominant). Observable fidelity is systematically better than pointwise, with $\Delta_{\text{obs}}/\Delta_{\text{ptw}}$ ranging from 0.018 (chemical-potential shift) to 0.41 (Compton y -distortion). The Teff recovery correctly captures the angular structure; the spectral error integrates out under energy weights. Among the 1008 scanned configurations, 742 fall in this regime.

Regime II (energy-shape distortions: $n \geq 2$ present). Observable fidelity is worse than pointwise, with $\Delta_{\text{obs}}/\Delta_{\text{ptw}}$ reaching 3.9 (quadratic distortion). The moment-matching optimiser absorbs energy-shape content into shifted Teff multipoles, generating spectral-angular cross-contamination. Among the scanned configurations, 266 fall in this regime.

The tangency diagnostic $\mathcal{D}_{\geq 2}$ of Paper I provides a physical predictor of which regime applies: when $\mathcal{D}_{\geq 2}$ is small, the off-manifold content is spectrally smooth (Regime I); when $\mathcal{D}_{\geq 2}$ is large, energy-shape distortions are present (Regime II).

V. COMPARISON WITH PRODUCTION CLOSURES

A. Setup

At a matched truncation order ($L = 2$), we compare five production closures: Levermore M_1 [7], Minerbo [10],

MEFD [26], PSTF [11], and Teff. The comparison is performed on four representative scenarios spanning the physically relevant closure-behaviour regimes: (i) quasi-thermal ($A = 0.05$, $Q = 0.02$): nearly isotropic, all closures near equilibrium; (ii) semi-transparent ($A = 0.15$, $Q = 0.10$): moderate anisotropy, typical of the neutri-nosphere region; (iii) beam-crossing ($A = 0.35$, $Q = -0.10$): configuration inaccessible to single-curve closures; (iv) strong anisotropy ($A = 0.45$, $Q = 0.20$): regime where closure differences are maximised.

B. Structural comparison

The five closures differ along six structural axes.

Realizability. Teff and MEFD guarantee $f \geq 0$ within their respective manifold domains ($\Theta > 0$ for Teff). Levermore M_1 is realisable by construction (single-curve maximum-entropy); Minerbo is realisable. PSTF truncation can produce negative occupation numbers at a fraction of phase-space points—at $(A, Q) = (0.45, 0.20)$, $|f_{\text{neg}}|_{\text{max}} \approx 6 \times 10^{-4}$ (Table III).

Closure dimensionality. Levermore M_1 , Minerbo, and MEFD map the flux factor $\mathbf{f}$ to a unique Eddington factor χ : the closure region is a one-dimensional curve in the $(\mathbf{f}, \chi)$ plane. PSTF and Teff are two-dimensional: they span a region $\chi(\mathbf{f}, Q)$ parameterised by both dipole and quadrupole. Beam-crossing configurations lie outside all single-curve closures; Teff and PSTF can represent them.

Spectral shape. Teff produces an equilibrium-shaped spectrum per ray ($f_{\Theta} = \Phi_{\xi}(y/\Theta)$); PSTF produces a polynomial perturbation of the angular dependence. The equilibrium shape is the natural assumption when collisions are efficient; the polynomial shape is appropriate when the spectral structure is resolved.

Moment exactness. PSTF is moment-exact by construction: the recovered moments $\ell \leq L$ match the true moments identically. Teff matches moments through a nonlinear inversion that is exact on-manifold but approximate off-manifold (Regime I/II of §IV D). Levermore M_1 and Minerbo match only the flux factor; the Eddington factor is a derived quantity.

Validity certificate. In the present comparison set, Teff is distinguished by an intrinsic tangency-based adequacy diagnostic ($\mathcal{D}_{\geq 2}$) that detects when the closure assumptions break down, with a quantitative bound on the observable error (Theorem 3). No other closure in this comparison offers a diagnostic-guided error estimate.

Interpretability. Teff provides a single per-direction number—the effective temperature $T_{\text{eff}}(\hat{e}) = T_0 \Theta(\hat{e})$ —that parameterises the full spectral shape. PSTF requires reassembling the multipole hierarchy $F_{\ell}(E) e^{A_{\ell}}$ to reconstruct the angular distribution. M_1 provides an Eddington factor but no per-direction spectral information.

C. Observable fidelity

On the quasi-thermal scenario, all five closures achieve $\Delta_{\text{obs}} < 10^{-3}$: near equilibrium, closure differences are negligible. On the semi-transparent scenario, Levermore M_1 and Minerbo agree with the reference to $\Delta_{\text{obs}} \sim 2 \times 10^{-2}$; PSTF and Teff to $\Delta_{\text{obs}} \sim 5 \times 10^{-3}$. On the beam-crossing scenario, M_1 and Minerbo produce $\Delta_{\text{obs}} \sim 10^{-1}$ (large Eddington-factor error); PSTF and Teff maintain $\Delta_{\text{obs}} \sim 2 \times 10^{-2}$. On the strong-anisotropy scenario, PSTF achieves the best observable fidelity ($\Delta_{\text{obs}} \approx 2.5 \times 10^{-3}$, moment-exact by construction); Teff is exact on-manifold ($\Delta_{\text{obs}} < 10^{-13}$) but off-manifold can reach $\Delta_{\text{obs}} \sim 4 \times 10^{-2}$ (Regime II).

The overall pattern: no single closure dominates uniformly. PSTF is marginally better on off-manifold distortions (moment exactness); Teff is exact on-manifold. M_1 fails at beam-crossing. Teff is the closure in this comparison that combines manifold-domain realizability ($\Theta > 0$) with a built-in diagnostic, at the cost of approximate (rather than exact) moment matching.

VI. SELF-CONSISTENT VALIDATION WITH BGK COLLISIONS

The off-manifold distortions of §III C are controlled perturbations imposed *a priori*. A more stringent test is whether the error bound (Theorem 3) holds for distributions generated self-consistently by a collision operator.

Using the Anderson–Witting BGK operator [27] $C[f] = (f_{\text{iso}} - f)/\tau$ with isotropisation timescale τ , we evolve anisotropic initial conditions ($A_0 = 0.30$) through six relaxation stages $\tau/\tau_0 \in \{0.1, 0.3, 0.5, 1.0, 2.0, 5.0\}$ for MB and FD statistics (12 configurations total). At each stage, the distribution has self-consistently generated spectral content beyond the Teff manifold; we measure the actual Δ_{obs} and compare with the bound $C_{\text{loc}} \mathcal{D}_{\geq 2}/\sigma_{\text{min}}(J)$.

Results. The bound is confirmed at all 12 configurations: the actual Δ_{obs} lies below the predicted bound. The bound is conservative by a factor of 10–30: the actual error is much smaller than the worst-case prediction. This conservatism is expected because Theorem 3 bounds the maximum over all distortion types, while the BGK operator generates a specific distortion pattern.

The diagnostic $\mathcal{D}_{\geq 2}$ decreases monotonically during isotropisation ($\mathcal{D}_{\geq 2} \propto \tau^{-1}$ at late times), tracking the relaxation of spectral content toward the manifold. At $\tau/\tau_0 = 5.0$, the distribution is nearly thermal and $\mathcal{D}_{\geq 2} < 10^{-3}$: the bound predicts $\Delta_{\text{obs}} < 0.003$, and the actual $\Delta_{\text{obs}} \approx 4 \times 10^{-4}$. The 2D diagnostic (Paper III, Theorem 3) further reduces this by absorbing the η -tangent component.

Significance. The BGK validation confirms that the error bound is not merely a mathematical artefact of the controlled distortion suite: it holds for physically generated spectra. Whether it holds equally well for more re-

alistic collision operators (neutrino–matter interactions, Compton scattering) remains to be tested.

VII. ASTROPHYSICAL IMPLICATIONS

A. Binary neutron star mergers

At typical post-merger anisotropy levels ($A \sim 0.1$ – 0.4) [28], the Teff $L = 2$ recovery of mean energy is controlled by $\mathcal{D}_{\geq 2}$: at $\mathcal{D}_{\geq 2} < 0.01$, $\delta\langle E_\nu \rangle / \langle E_\nu \rangle \lesssim 2\%$, subdominant to the 10–30% M_1 –MC discrepancy [1]. Since $\sigma_{\nu N} \propto E_\nu^2$ [25], a 3% mean energy bias propagates to $\delta Y_e \sim 0.01$ – 0.02 . The M_1 radiation trapping problem [2] does not arise in Teff because the (f, χ) closure region is two-dimensional.

B. CMB spectral distortions

For CMB photons (Planck statistics), $T_{\text{eff}}(\hat{e})$ is the exact brightness temperature. The temperature-distribution parameterisation of Barenboim et al. [29] is the spectral-domain analogue; the Teff framework additionally provides $\mathcal{D}_{\geq 2}$ and Theorem 3. The generalised Boltzmann hierarchy [30] provides the spectral degrees of freedom that Teff angular multipoles complement.

C. Core-collapse supernovae

The nonmonotonicity of Wang & Burrows [5] applies to all closures: local adequacy is necessary but not sufficient for dynamical adequacy. Teff provides the computable diagnostic that other closures in this comparison lack. The fast-flavor conversion results of Akaho et al. [31] suggest that closure accuracy in the semi-transparent region can bifurcate explosion outcomes, making a diagnostic bridge practically relevant.

VIII. KNOWN LIMITS

- (i) Spatial transport not coupled.
- (ii) Flavor mixing not treated [31].
- (iii) Pair processes not included.
- (iv) Observable set not exhaustive.
- (v) Axisymmetric $L = 2$ only.
- (vi) Theorem 3 constants numerically determined, not analytically derived.
- (vii) Realizability holds within the manifold domain ($\Theta > 0$); configurations violating this bound are outside the ansatz.

TABLE I. Temperature bridge relation and statistics independence. T_{bol}/T_0 (exact) vs perturbative prediction (3). Flux factor and Eddington factor verified identical for MB, FD, and BE to $< 6 \times 10^{-17}$ at all entries.

A	Q	T_{bol}/T_0 (exact)	T_{bol}/T_0 (pert.)	T_{num}/T_0
0.05	0	1.001248	1.001250	1.000833
0.10	0	1.004968	1.005000	1.003322
0.20	0	1.019502	1.020000	1.013159
0.30	0	1.042604	1.045000	1.029142
0.20	0.10	1.021802	—	1.014376
0.30	0.15	1.048554	—	1.032173

TABLE II. Off-family recovery: adequacy metrics. True angular structure $(A, Q) = (0.25, 0.12)$. Regime I: spectral distortions ($n \leq 1$); Regime II: energy-shape distortions ($n \geq 2$).

Distortion	Δ_{obs}	Δ_{func}	Δ_{ptw}	Obs/Ptw	Regime
y (0.01)	8.6×10^{-3}	1.9×10^{-2}	2.1×10^{-2}	0.41	I
y (0.05)	4.4×10^{-2}	1.0×10^{-1}	1.2×10^{-1}	0.38	I
μ (0.01)	1.2×10^{-2}	7.5×10^{-2}	6.6×10^{-1}	0.018	I
μ (0.05)	6.1×10^{-2}	2.4×10^{-1}	3.3	0.019	I
quad (0.005)	6.6×10^{-2}	4.5×10^{-2}	1.7×10^{-2}	3.9	II
quad (0.02)	1.9×10^{-1}	1.7×10^{-1}	6.8×10^{-2}	2.7	II

IX. DISCUSSION AND OUTLOOK

For the neutrino transport community, Theorem 3 provides a conditional local bound on the observable reconstruction error, enabling a hybrid strategy: Teff in the diagnostic-certified regime, Monte Carlo when $\mathcal{D}_{\geq 2}$ exceeds a user-specified threshold. For the CMB community, Teff preserves exact brightness temperature for

blackbody photons ($\mu = 0$ Planck distribution) and yields high observable fidelity under spectral distortions. For the closure theory community, no closure simultaneously dominates in accuracy, realizability, and interpretability; Teff combines manifold-domain realizability with a built-in diagnostic, at the cost of approximate (rather than exact) moment matching.

Three extensions follow: (i) quantum-closure coupling [15, 16]; (ii) spectro-angular temperature distribution combining the angular Teff parameterisation with spectral parameters [29, 30]; (iii) validation on Monte Carlo-generated spectra from post-merger environments.

Because the closure classes compared here are not nested, the present comparison does not establish universal pointwise dominance of Teff over all alternatives; it should be read as a structural empirical comparison at matched angular budget.

TABLE III. Structural property comparison: Teff vs PSTF.

Property	Teff	PSTF
Realizability ($f \geq 0$)	Yes, when $\Theta > 0$	Not guaranteed
Spectral shape	Equilibrium	Polynomial
Moment exactness ($\ell \leq L$)	Approximate	Exact
High- ℓ tail ($\ell > L$)	Nonlinear tail	Zero
Interpretability	$T_{\text{eff}}(\hat{e})$ readable	Reassembly needed
Validity diagnostic	$\mathcal{D}_{\geq 2}$ (conditional)	None

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Data availability. The Python code is available as Supplemental Material [URL will be inserted by publisher].

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